



Started on	Monday, 31 August 2026, 8:13 AM
State	Finished
Completed on	Monday, 31 August 2026, 8:17 AM
Time taken	3 mins 37 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int countZeroes(int arr[], int low, int high, int n)
4  {
5      if (low > high)
6          return 0;
7      int mid = (low + high) / 2;
8      if (arr[mid] == 0 && (mid == 0 || arr[mid - 1] == 1))
9      {
10         return n - mid;
11     }
12     if (arr[mid] == 1)
13     {
14         return countZeroes(arr, mid + 1, high, n);
15     }
16     return countZeroes(arr, low, mid - 1, n);
17 }
18
19 int main()
20 {
21     int m;
22     scanf("%d", &m);
23
24     int arr[m];
25
26     for (int i = 0; i < m; i++)
27     {
28         scanf("%d", &arr[i]);
29     }
30
31     printf("%d", countZeroes(arr, 0, m - 1, m));
32
33     return 0;
34 }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Charunetra R 250701123-2025-CSE ▾

C2

Started on	Monday, 31 August 2026, 11:20 AM
State	Finished
Completed on	Monday, 31 August 2026, 11:41 AM
Time taken	21 mins 27 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: `3`

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: `2`

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  int count(int arr[], int left, int right, int value)
3  {
4      int i, c = 0;
5      for (i = left; i <= right; i++)
6      {
7          if (arr[i] == value)
8              c++;
9      }
10     return c;
11 }
12 int majorityElement(int arr[], int left, int right)
13 {
14     if (left == right)
15         return arr[left];
16     int mid = left + (right - left) / 2;
17     int leftMajority = majorityElement(arr, left, mid);
18     int rightMajority = majorityElement(arr, mid + 1, right);
19     if (leftMajority == rightMajority)
20         return leftMajority;
21     int leftCount = count(arr, left, right, leftMajority);
22     int rightCount = count(arr, left, right, rightMajority);
23
24     if (leftCount > rightCount)
25         return leftMajority;
26     else
27         return rightMajority;
28 }
```

```
27         return rightMajority;
28     }
29     int main()
30     {
31         int n;
32         scanf("%d", &n);
33         int nums[n];
34         for (int i = 0; i < n; i++)
35         {
36             scanf("%d", &nums[i]);
37         }
38         printf("%d", majorityElement(nums, 0, n - 1));
39         return 0;
40     }
```

	Input	Expected	Got	
✓	3	3	3	✓
	3 2 3			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 31 August 2026, 8:22 AM
State	Finished
Completed on	Monday, 31 August 2026, 8:26 AM
Time taken	3 mins 54 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int findFloor(int arr[], int low, int high, int x)
3 {
4     if (low > high)
5         return -1;
6     int mid = low + (high - low) / 2;
7     if (arr[mid] == x)
8         return arr[mid];
9     if (arr[mid] < x)
10    {
11        int result = findFloor(arr, mid + 1, high, x);
12        if (result == -1)
13            return arr[mid];
14        return result;
15    }
16    return findFloor(arr, low, mid - 1, x);
17 }
18 int main()
19 {
20     int n, x;
21     scanf("%d", &n);
22     int arr[n];
23     for (int i = 0; i < n; i++)
24     {
25         scanf("%d", &arr[i]);
26     }
27     scanf("%d", &x);
28     int result = findFloor(arr, 0, n - 1, x);
29     printf("%d", result);
30     return 0;
31 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			

	Input	Expected	Got	
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 31 August 2026, 8:26 AM
State	Finished
Completed on	Monday, 31 August 2026, 8:29 AM
Time taken	2 mins 5 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int findPair(int arr[], int left, int right, int x, int *a, int *b)
3  {
4      if (left >= right)
5          return 0;
6      int sum = arr[left] + arr[right];
7      if (sum == x)
8      {
9          *a = arr[left];
10         *b = arr[right];
11         return 1;
12     }
13     if (sum < x)
14         return findPair(arr, left + 1, right, x, a, b);
15     return findPair(arr, left, right - 1, x, a, b);
16 }
17 int main()
18 {
19     int n, x;
20     scanf("%d", &n);
21     int arr[n];
22     for (int i = 0; i < n; i++)
23     {
24         scanf("%d", &arr[i]);
25     }
26     scanf("%d", &x);
27     int a, b;
28     if (findPair(arr, 0, n - 1, x, &a, &b))
29     {
30         printf("%d\n", a);
31         printf("%d\n", b);
32     }
33     else
34     {
35         printf("No");
36     }
37     return 0;
38 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 31 August 2026, 8:29 AM
State	Finished
Completed on	Monday, 31 August 2026, 8:33 AM
Time taken	3 mins 16 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2  void swap(int *a, int *b)
3  {
4      int temp = *a;
5      *a = *b;
6      *b = temp;
7  }
8  int partition(int arr[], int low, int high)
9  {
10     int pivot = arr[high];
11     int i = low - 1;
12     for (int j = low; j < high; j++)
13     {
14         if (arr[j] <= pivot)
15         {
16             i++;
17             swap(&arr[i], &arr[j]);
18         }
19     }
20     swap(&arr[i + 1], &arr[high]);
21     return i + 1;
22 }
23 void quickSort(int arr[], int low, int high)
24 {
25     if (low < high)
26     {
27         int pi = partition(arr, low, high);
28         quickSort(arr, low, pi - 1);
29         quickSort(arr, pi + 1, high);
30     }
31 }
32 int main()
33 {
34     int n;
35     scanf("%d", &n);
36     int arr[n];
37     for (int i = 0; i < n; i++)
38     {
39         scanf("%d", &arr[i]);
40     }
41     quickSort(arr, 0, n - 1);
42     for (int i = 0; i < n; i++)
43     {
44         printf("%d", arr[i]);
45         if (i < n - 1)
46             printf(" ");

```

```
47 |     }  
48 |     return 0;  
49 | }
```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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